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In each question, quantified variables have domain $\mathbb{Z}$ unless otherwise specified.

1. Remove redundant parentheses for the following propositions and predicates.
 - a. $(p \rightarrow (p \vee q)) \rightarrow q$
 - b. $((p \leftrightarrow q) \rightarrow (r \leftrightarrow p))$
 - c. $(p \vee ((q \wedge (r \vee p)) \wedge r))$
 - d. $\forall x \in \mathbb{Z}. ((x > 0) \rightarrow (\exists y. (y < x)))$
 - e. $(\forall x > 0. (\forall y > 0. (x * y = 1))) \rightarrow ((x = y) \wedge (x = 1))$
2. Use Truth Tables to prove the following logical equivalences.
 - a. $(p \vee q) \wedge q \Leftrightarrow q$
 - b. $\neg(p \leftrightarrow q) \Leftrightarrow \neg p \leftrightarrow q$
 - c. $\neg p \wedge (p \vee q) \rightarrow q \Leftrightarrow T$
3. Give formal proof of truth for each of the following logical equivalences using only rules provided in Lecture 2. Remember to clarify which rule(s) you use in each step.
 - a. $(p \vee q) \wedge \neg q \Leftrightarrow \neg(p \rightarrow q)$
 - b. $\neg(p \wedge q) \Leftrightarrow \neg q \vee q \wedge \neg p$
 - c. $q \Leftrightarrow (p \rightarrow q) \wedge (\neg p \rightarrow q)$
4. Give formal proof of truth for each of the following logical implications using only rules provided in Lecture 2. Remember to clarify which rule(s) you use in each step.
 - a. $\neg(p \wedge q) \wedge p \Rightarrow \neg q$
 - b. $p \wedge q \vee q \wedge r \Rightarrow p \vee q \vee r$
 - c. $(p \rightarrow q) \wedge (\neg p \rightarrow r) \Rightarrow q \vee r$ Hint: Try to conjunct $(p \vee \neg p)$ to the left-hand side
5. Decide true or false for each of the following statements about states, satisfaction or validity. Briefly explain your answers.
 - a. $\{x = 2, p = T\} \models x > 2 \vee p$.
 - b. $\{b = 5, i = 0, x = 6\}$ is proper for predicate $x > b[i]$.
 - c. Let $\sigma = \{p = F, b = (2, 0, 4)\}$, then $\sigma \models p \leftrightarrow b[b[0]] = 2$.
 - d. $\not\models p \rightarrow p \vee q$
 - e. $\emptyset \models T$
6. Give formal proof of truth for each of the following logical equivalences using rules provided in Lecture 2 and 3. Remember to clarify which rule(s) you use in each step.
 - a. $\neg \forall x \geq 1. x^2 > x \Leftrightarrow \exists x. x \geq 1 \wedge x^2 \leq x$
 - b. $\neg \exists x. \exists y. x > y \wedge x < y \Leftrightarrow \forall x. \forall y. x \leq y \vee x \geq y$

- c. $\neg((\exists x. \exists y. Q(x, y)) \wedge \forall x. \forall y. Q(y, x)) \Leftrightarrow (\forall x. \forall y. \neg Q(x, y)) \vee \exists x. \exists y. \neg Q(y, x)$ Here, $Q(x, y)$ is a predicate function.

7. Define the following predicate functions.

- a. Define predicate function *isGreater*(b, m, x) which returns *True* if and only if positive m is not larger than the length of array b , and each of the first m numbers in b is greater than x . For example, *isGreater*($(3, 4, 6, 1), 3, 2$) returns *True*, *isGreater*($(3, 4, 6, 1), 3, 4$) returns *False*.
- b. Define predicate function *hasGreater*(a, b) which returns *True* if and only if every integer in array a is greater than some integer in array b . For example, *hasGreater*($(4, 5, 3, 8), (3, 2, 5)$) returns *True*, *hasGreater*($(4, 5, 3, 8), (3, 4, 5)$) returns *False*.
- c. Define predicate function *Extends*(a, b) which returns *True* if and only if array b is an extension of array a ; in other words, we have $a[0] = b[0]$, $a[1] = b[1]$, $a[2] = b[2]$... for all elements in a . For example, *Extends*($(1, 4, 2), (1, 4, 2, 9, 3)$) returns *True*.

8. Let e_1 and e_2 be two expressions.

- a. In general, does $e_1 = e_2$ logically imply $e_1 \equiv e_2$? If yes, briefly justify it (in a sentence or two); if not, give a counterexample.
- b. In general, does $e_1 \neq e_2$ logically imply $e_1 \not\equiv e_2$? If yes, briefly justify it (in a sentence or two); if not, give a counterexample.